

概率统计 22-23-2(A)标准答案及评分标准

一、选择题

1)C 2)C 3)C 4)B, 5)C

二、填空题

1)11/21=0.5238; 2)0.0243; 3)12; 4)25/36; 5)19.6; 6) $-\frac{8}{5\sqrt{30}}$; 7) 13/4=3.25; 8) 0.048

$$9) F(x) = \begin{cases} 0 & x < -3 \\ 0.4 & -3 \leq x < 1 \\ 0.7 & 1 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

$$10) f_Y(y) = \begin{cases} \frac{0.375}{\sqrt{y}} & 0 < y < 1 \\ \frac{0.125}{\sqrt{y}} & 1 \leq y < 4 \\ 0 & \text{其它} \end{cases}$$

11) $\chi^2(4)$; 12) [8.04, 11.96]; 13) 0.4

三、

$$\text{三、 (1)} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1;$$

$$a \int_{-1}^1 \int_{x^2}^{x^2+1} dy dx = 1;$$

$$a = \frac{1}{2}$$

$$(2) f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$\text{当 } 0 < y < 1 \text{ 时 } f_Y(y) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \sqrt{y}$$

$$\text{当 } 1 \leq y < 2 \text{ 时, } f_Y(y) = 2 \int_{\sqrt{y-1}}^1 \frac{1}{2} dx = 1 - \sqrt{y-1}$$

$$(3) f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \begin{cases} \frac{1}{2\sqrt{y}} & x^2 < y. \\ 0 & \text{其他} \end{cases} \quad (0 < y < 1)$$

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \begin{cases} \frac{1}{2(1 - \sqrt{y-1})} & x^2 < y < x^2 + 1. \\ 0 & \text{其他} \end{cases} \quad (1 < y < 2).$$

$$f_{X|Y}(x|0.25) = \begin{cases} 1 & |x| < \frac{1}{2} \\ 0 & \text{其它} \end{cases}$$

$$P(X < 0.3 | Y = 0.25) = \int_{-\infty}^{0.3} f_{X|Y}(x|0.25) dx = \int_{-0.5}^{0.3} 1 dx = 0.8$$

四、 A_i 表示有*i*个元件发生故障, $i = 0,1,2,3$; B 表示电子产品发生故障.

$$P(A_0) = 0.8 \times 0.7 \times 0.8 = 0.448;$$

$$P(A_1) = 0.2 \times 0.7 \times 0.8 + 0.8 \times 0.3 \times 0.8 + 0.8 \times 0.7 \times 0.2 = 0.416;$$

$$P(A_2) = 0.2 \times 0.3 \times 0.8 + 0.2 \times 0.7 \times 0.2 + 0.8 \times 0.3 \times 0.2 = 0.124;$$

$$P(A_3) = 0.2 \times 0.3 \times 0.2 = 0.012;$$

$$P(B|A_0) = 0; P(B|A_1) = 0.3; P(B|A_2) = 0.6; P(B|A_3) = 1$$

$$(1) P(B) = \sum_{i=0}^3 P(A_i)P(B|A_i)$$

$$= 0.448 \times 0 + 0.416 \times 0.3 + 0.124 \times 0.6 + 0.012 \times 1 = 0.2112$$

(2)

$$\begin{aligned} P(A_1|B) &= \frac{P(A_1)P(B|A_1)}{P(B)} \\ &= \frac{0.416 \times 0.3}{0.2112} = 0.591 \end{aligned}$$

五、 Z 的分布函数 $F_Z(z) = P(Z \leq z) = P(XY \leq z)$

当 $z < 0$ 时, $F_Z(z) = 0$

当 $z > 1/4$ 时, $F_Z(z) = 1$

$$\begin{cases} x + y = 1 \\ xy = z \end{cases}; \text{解得: } x_1 = \frac{1}{2}(1 - \sqrt{1 - 4z}); x_2 = \frac{1}{2}(1 + \sqrt{1 - 4z})$$

$$\text{当 } 0 \leq z \leq \frac{1}{4} \text{ 时, } F_Z(z) = \iint_{xy \leq z} f(x, y) dx dy$$

$$= 1 - 2 \int_{x_1}^{x_2} \int_{z/x}^{1-x} dy dx$$

$$= 1 - \sqrt{1 - 4z} + 4z \ln(1 + \sqrt{1 - 4z}) - 2z \ln 4z$$

Z 的概率密度为

$$f_Z(z) = [F_Z(z)]' = \begin{cases} 4 \ln(1 + \sqrt{1 - 4z}) - 2 \ln 4z & 0 < z \leq \frac{1}{4}, \\ 0 & \text{其它} \end{cases}$$

$$\text{方法 2: } f_Z(z) = \int_{-\infty}^{+\infty} \frac{1}{|x|} f\left(x, \frac{z}{x}\right) dx$$

当 $0 < z < 1/4$ 时

$$0 < x < 1, x + \frac{z}{x} < 1; \quad x_1 < x < x_2, x_1 = \frac{1}{2}(1 - \sqrt{1 - 4z}); x_2 = \frac{1}{2}(1 + \sqrt{1 - 4z})$$

$$f_Z(z) = \int_{x_1}^{x_2} \frac{2}{|x|} dx = 2 \ln\left(\frac{x_2}{x_1}\right) = 4 \ln(1 + \sqrt{1 - 4z}) - 2 \ln 4z;$$

当 $z < 0$, or $z > \frac{1}{4}$

$$f_Z(z) = 0,$$

六、 X_i 表示第 i 次抽出球的号数， $i = 1, 2, \dots, 100$. 则 X_i 的分布为， $P(X_i = 1) = \frac{1}{6}$; $P(X_i = 2) = \frac{1}{3}$; $P(X_i = 3) = \frac{1}{2}$; $EX_i = \frac{7}{3}$; $DX_i = \frac{5}{9}$

所求概率为:

$$P\left(\sum_{i=1}^{100} X_i \leq 210\right) \approx \Phi\left(\frac{210 - 100 \times \frac{7}{3}}{\sqrt{100 \times \frac{5}{9}}}\right) \\ = \Phi\left(\frac{-7}{\sqrt{5}}\right) = 1 - \Phi(3.13)$$

七、(1) 似然函数为: $L(\theta) = \prod_{i=1}^n f(X_i) = \prod_{i=1}^n \theta X_i^{-2}$.

$$= \theta^n \prod_{i=1}^n X_i^{-2}, \theta \leq \min(X_1, \dots, X_n).$$

$L(\theta)$ 是 θ 的单调增函数，所以，当 $\theta = \min(X_1, \dots, X_n)$ 时， $L(\theta)$ 取最大值。
故 θ 的最大似然估计量为:

$$\hat{\theta} = \min(X_1, \dots, X_n)$$

(2) 总体的分布函数为:

$$F(x) = \begin{cases} 1 - \frac{\theta}{x} & x \geq \theta \\ 0 & x < \theta \end{cases} \quad \text{A}$$

$\hat{\theta}$ 的概密度函数为， $F_{\hat{\theta}}(t) = n[F(t)]^{n-1}f(t)$

$$= \begin{cases} n\theta^n t^{-(n+1)} & t \geq \theta \\ 0 & t < \theta \end{cases}$$

$$E\hat{\theta} = \int_{\theta}^{+\infty} tn\theta^n t^{-(n+1)} dt = \frac{n}{n-1}\theta$$

$E\hat{\theta} \neq \theta$, $\hat{\theta}$ 不是 θ 的无偏估计

八、(1) $n = 25, \alpha = 0.05$,

$$\text{检验统计量 } T = \frac{\bar{X} - 54}{S_n} \sqrt{n} | H_0 \sim t(n-1)$$

$$\text{拒绝域: } D = \{T < -t_{\alpha}(n-1)\} = \{T < -1.7113\}$$

$$\bar{x} = 52, s_n = 3$$

$$T \text{ 的观测值: } T = \frac{52 - 54}{3} \sqrt{25} = -3.33$$

$$-3.33 < -1.7113,$$

所以，拒绝原假设。

$$(2) \sigma^2 \text{ 的置信度为 } 95\% \text{ 的置信区间为: } \left[\frac{(n-1)S_n^2}{\chi_{0.025}^2(24)}, \frac{(n-1)S_n^2}{\chi_{0.975}^2(24)} \right]$$

$$= \left[\frac{24 \times 9}{39.36}, \frac{24 \times 9}{12.4} \right] = [5.488, 17.419]$$

所以， σ 的置信度为 95% 的置信区间为

$$[\sqrt{5.488}, \sqrt{17.419}] = [2.343, 4.174].$$